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A demographic comparison and characterization of pediatric poisoning before and after the emergence of COVID-19

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ABSTRACT

Background: To compare relative rates of pediatric poisoning before and after COVID-19, including by demographic and urban-rural status, and by agent identified, using data from one university healthcare system and children's hospital.

Methods: Using retrospective, cross sectional design from deidentified healthcare claims data, we extracted all encounters with the *ICD-10-CM for Poisoning by, Adverse effects of, and Underdosing of drugs, medicants and biological substances (T36-T50)* and grouped the encounters as those after state mandates regulating activity came into effect (Post-COVID-19 (3/17/2020–3/18/2021)) Pre-COVID-19 (3/18/2019–3/17/2020). We then compared poisoning agent, age at the time of the encounter, recorded sex, race, ethnicity, rural/urban residence, and visit type using Mann-Whitney *U* test, chi-square test of association, incidence rates and incident rate ratios between the time periods.

Findings: The sample included 1608 unique patients 0–17 years of age and 4216 encounters. We also identified IRRs >1 in nearly every demographic subgroup with the exception of Non-Hispanic Blacks. The comparison of specific drugs or medicants identified a significant decrease in poisoning by *Systemic antibiotics* (T36); but an increase in *Hormones and their synthetic substitutes and antagonists* (T38), *Non opioid analgesics antipyretic and antirheumatic* (T39), *Psychotropic Drugs* (T39) and *Systemic and hematologic agents* (T45).

Conclusion: This study identifies pediatric subgroups highly affected by pediatric poisoning during the time-period immediately after the identification of COVID-19 and characterizes the drugs commonly associated with poisonings.

Application to practice: With a further understanding nursing has the potential to impact pediatric poisoning in the inpatient, outpatient and public health setting.

Introduction

Pediatric poisonings have been identified as a one of six primary sources of injury in U.S. children targeted by the Center for Disease Control and Prevention (Center for Disease Control and Prevention, 2021). Of the six injury indications, pediatric poisoning cases have demonstrated the largest yearly increase (Center for Disease Control and Prevention, 2021). In 2020, 58,200 children were injured by accidental pediatric poisoning in the U.S. and 43 children died as a result (U.S. Consumer Product Safety Commission, 2022). Deaths occur disproportionately for those under the age of 5 with 100% increase in the number of deaths in this age group from 2018 to 2019; there was an increase of 26% from 2019 to 2020, the largest annual increase in recent years (U.S. Consumer Product Safety Commission, 2022). To decrease rates and prevent child morbidity and mortality, it is critically important that high-risk groups be identified to develop effective targeted, developmentally appropriate approaches.

The American Association of Poison Control Center reported that in 2020 there were 1.2 million poison exposures to children (< 20 years), and 92% occurred in the residence. As such, lack of safe

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medication storage in the home and appropriate parental supervision are a key environmental risk factors for pediatric poisoning (Gummin et al., 2021; WHO Guidelines Approved by the Guidelines Review Committee, 2008).

Due to the emergence of COVID-19 pandemic, a Federal public health state of emergency was declared and state stay-at-home mandates were ordered (Center for Disease Control and Prevention, 2023a). As a resulted, families across the U.S. spent extended periods of time in their residence. Recognizing that homes are the primary location for pediatric poisoning and more time was spent in homes because of COVID-19, one would postulate that pediatric poisoning would increase in the U.S. during the time period affected by COVID-19. Yet, conflicting findings are reported in the literature. Specifically, a 6.3% decrease in pediatric poisoning was reported using data from the American Association of Poison Control Centers National Poison Data System (Lelak et al., 2021). Using emergency department data, an increase in pediatric poisoning visits were reported in Canada ($p < .001$) (Zhang et al., 2022). In retrospective U.S. cohort study investigating mortality and mortality risk, no difference in rates of pediatric poisoning were identified ($p = .86$) (Williams et al., 2021).

It is critically important that we understand the implications of the COVID-19 pandemic on the health of populations for several reasons. First, we understand the pandemic systematically changed human behaviors and these changes have persisted during despite the elimination of governmental restrictions (Yabe et al., 2023). One such example of changes in behaviors is the continued increase in the number of children homeschooled (2019–2.8% of children in the U.S. vs. 5.4% in 2023) (Reason Foundation, 2023). The risks to children are undeniably different in these two environments and should be considered by those who aim to address health outcomes. Second, the effects of COVID-19 on the health of populations continues despite the acuity of the infectious aspects of the condition lessening in impact (Substance Abuse and Mental Health Services Administration, 2023). A full characterization of the effects of COVID-19 on pediatric poisoning is needed to provide healthcare providers with the guidance to the current state of the condition in the U.S. It can also make clear the risk to those children who continue in daily living situations that closely parallel that seen during the COVID-19 pandemic.

Due to the conflicting findings reported in the literature and, generally, a lack of full characterization of the effects of COVID-19 on the prevalence of pediatric poisoning in the U.S; specifically, demographic comparisons across health environments (i.e., inpatient, outpatient and emergency department) is a significant gap along with a consideration of drug classifications. It is imperative that we understand the implications of the pandemic on pediatric poisoning, to prevent child injury. The identification of disparities in rates of pediatric poisoning can facilitate a targeted approach to provide services and resources to high need populations.

Purpose

The purpose of this project was to use healthcare claims data to evaluate pediatric poisonings in those <18 years of age seen at one university healthcare system and children's hospital serving Central, Eastern, and Southern counties in one U.S. state. We compared rates of pediatric poisoning between the dates of March 18, 2020-March 17, 2021 to March 18, 2019-March 17, 2020, and evaluated whether the distribution of cases was associated with urban-rural residence of or class of substance identified. We hypothesized that because nearly all human exposures to poison occur with in the residence (92%), children may have been more at risk for pediatric poisoning encounters during the COVID-19 affected time period, given the state-wide stay-at-home mandates during the time, relative to the immediate pre-pandemic time period.(Commonwealth of Kentucky, 2023; Gummin et al., 2021).

Methods

This study was a retrospective, cross sectional study which utilized deidentified claims data that included demographic indicators. The study utilized one U.S. public university's Clinical and Translational Science bioinformatics services as honest brokers to obtain deidentified claims data. All billed patient encounters with the code for *International Classification of Diseases ICD-10-CM for Poisoning by, Adverse effects of, and Underdosing of drugs, medicants and biological substances (T36-T50)* with the dates of March 18, 2020-March 17, 2021, or to March 18, 2019 to March 17, 2020 who were < 18 years of age were included in the sample (Sawyer et al., 2019). Those with qualifying encounters between the dates of March 18, 2019 to March 17, 2020 were grouped in the Pre-COVID-19 cohort. Those encounters from March 18, 2020 to March 18, 2021, the time period immediately following the start of COVID-19 pandemic, were grouped in the Post-COVID-19 cohort. The U.S. state's Executive Orders declaring State of Emergency guided the date used to distinguish between cohorts (BeShear, 2020). For the most prevalent code (T45- Systemic and hematologic agents, not otherwise specified), descriptive statistics were used to describe the most specific ICD-10 codes (e.g., T45. 1X5A).

Extracted data included the substance that resulted in the poisoning, 'age at the time of the encounter', 'recorded sex', 'race', 'ethnicity', 'Rural Urban Continuum Codes' (RUC), and 'visit type' (Inpatient, Outpatient or Emergency Department) (USDA United States Department of Agriculture Economic Research Service). Age group categories were designated based on the literature and developmental milestones, specifically considering gross and fine motor development along behavioral characteristics of age group categories (Center for Disease Control and Prevention, 2023b; Hardin & Hackell, 2017; Sawyer et al., 2019). Participants were grouped into rural versus urban groups based on their residence using RUC codes (USDA United States Department of Agriculture Economic Research Service, 2024).

Data analysis

Demographic characteristics of unique patients were summarized using frequency distributions. Cohort differences (i.e., Pre- versus Post-COVID-19 periods) in demographic variables were evaluated using the Mann-Whitney U test or chi-square test of association, as appropriate. Incidence rates (IRs) were calculated to estimate rate of pediatric poisoning between the two time cohorts. To account for health care utilization frequency in the Pre- and the Post-COVID-19 time periods, we used proportions with total healthcare encounters (referred to as 'total encounters' in result tables) in pediatric patients among Pre- and Post-COVID-19, respectively, as the denominators. For some analyses the numerator was the aggregated ICD-10 codes for pediatric poisoning encounters (T36-T50) and others the ICD-10 code independently (e.g., T36). This analysis allowed for a comparison of rates before and during the pandemic that adjusts for the overall decline in healthcare utilization during the time period immediately following the widespread emergence of COVID-19. This approach to the consideration of healthcare utilization during COVID-19 is support in the literature (Salt et al., 2021; Swedo et al., 2020). Changes in IRs were evaluated in pediatric encounters in the Pre- and Post-COVID-19 time periods by demographic groups in the total, Pre- and Post- samples using incidence rate ratios (IRRs) to make the comparisons. IRRs and corresponding 95% confidence intervals and p-values were calculated using OpenEpi (Sullivan et al., 2009). We also used IRRs to evaluate differences in ICD-10 codes Pre- and Post-COVID-19 within each age group. All other data analysis was conducted using SAS©, version 9.4 (Cary, NC). An alpha of 0.05 was used to determine statistical significance for inferential testing.

Ethical considerations

This study was approved by a medical institutional review board (#45668) at one public university. The data analyzed was deidentified.

Results

Our data analysis considered both *encounters* (also referred to as claims) and *unique patients* that had the targeted ICD-10 codes. In the two-year study time-period (March 2019 to March 2021), there were 4216 encounters that involved 1608 unique patients 0–17 years of age. Nearly one-quarter of unique patients ($n = 384$, or 24%) had more than one encounter during the study period. We considered the codes associated with pediatric poisoning in aggregate (ICD-10: T36-T50; Table 1) and by drug categories using the ICD-10 specific to drug or medication (e.g., T36, T37; Table 3). There were 15 ICD-10 codes with 10 or more encounters per code in the Pre- and Post-COVID time periods. We do not present ICD-10 diagnosis codes with <10 encounters across the study period within specific age groups, or those that were non-significant for all age-specific comparisons (i.e., ICD-10 codes T44, T46, T48 and T49).

Demographically, there were 873 unique patients identified in the Pre-COVID-19 and 735 in the Post-COVID-19 time period. Across the study period, the majority of the sample was white, non-Hispanic (82%) and nearly half resided in a rural area (47%). Among the patients who met inclusion criteria, those aged 0–4 (36%) and 14–17 (31%) accounted for the majority of the sample. Slightly over half (53%) were female (Table 1).

In our comparison of the two time-periods, there were no differences between the Pre- and Post-COVID-19 cohorts in urban/rural county of residence or race/ethnicity. Interestingly, we did identify significant differences in the age distribution between the Pre- and Post-COVID time periods ($p = .002$) (Fig. 1). In the Pre-COVID-19 time period, there was a larger percentage of the sample with an age younger than 5 years (Pre-COVID-19 = 38% vs. Post-COVID-19 = 33%). Conversely, in the Post-COVID-19 period there was a larger percentage of the sample that was aged 14–17 (Post-COVID-19 = 34% vs. Pre-COVID-19 = 29%).

In the following the IRRs of the overall rate of pediatric poisoning by each specific demographic group are reported. There was a decrease in the overall number of encounters in Post-COVID-19 time period, Yet, using incident rate ratios, which accounts for the decreased healthcare utilization observed during the COVID-19 pandemic, there was a 41% increase in the rate of pediatric poisoning encounters from the Pre- to Post-COVID period ($IRR = 1.41$, 95% $CI = 1.33–1.50$; $p < .001$; see Table 2). There was a significant increase in the rate of encounters for all age groups (ages 0–4 = 26% increase [$p < .001$]; ages 5–10 = 55% increase [$p < .001$]; ages 11–13 = 20% increase [$p < .03$]; and ages 13–17 = 61% increase [$p < .001$]). Similarly, increases in rates were identified in those who were urban dwelling (28% increase, $p < .001$) and rural-dwelling (52% increase, $p < .001$). Although all races experienced increased rates in the Post-COVID-19 period, the highest rate increase was among white, non-Hispanics ($IRR = 1.46$; $p < .001$; Table 2). Findings regarding age and encounter setting (e.g., inpatient, outpatient, emergency department) are described in Table 2.

In the analysis using IRR using specific ICD-10 codes for drug classification (e.g., T36, T37), there was a 66% increase in the rate of encounters for *Poisoning by systemic and hematologic agents not otherwise specified* [T45] ($p < .001$) in the Post- versus the Pre-COVID-19 time period (Table 3). There was a 48% increase in encounters for *Hormones and their synthetic substitutes and antagonists* (T38; $p = .026$); a 47% increase in *psychotropic drugs* (T38; $p = .003$). There was a nearly significant increase in *Non-opioid analgesics antipyretics substitutes and antirheumatic* (T39; $IRR = 1.31$; $p = .052$), and a significant decrease in the rate of encounters due to the *Poisoning by systemic antibiotics* (T36, $IRR = 0.50$; $p < .001$).

The following data reports the results from the evaluation of ICD-10 diagnosis codes associated with pediatric poisoning claims by age group related changes between the Pre- vs. Post-COVID-19 periods and complete data is reported in Table 4. A significant decrease in the rate of encounters due *to Poisoning by systemic antibiotics* (T36, $IRR = 0.51$, $p < .001$, see Table 4) was also found in this age group. For ages 5–10, there were significant increases in claims due to *Poisoning by systemic and hematologic agents not otherwise specified* (T45, $IRR = 1.90$, $p < .001$) and *Poisoning by systemic antibiotics* (T36, $IRR = 1.65$, $p < .001$). A decrease in the rate of claims due to *Diuretics and other unspecified drugs, medicants and biologic substances* (T50, $IRR = 0.58$, $p = .042$) was also identified in this age group.

The following text describes the most specific classification of the ICD-10 code for *Poisoning by systemic and hematologic agents not otherwise specified,* the most prevalent code. In the Pre-COVID-19 time-period 97% ($n = 1067$) of the encounters were for ICD-10 codes T45.1X5A, T45.1X5D, or T45.1X5S which is described as *Adverse effect of an antineoplastic and immunosuppressive drug, initial encounter,*

Table 1
Demographic summary of pediatric poisoning for unique patients by study period.

	Total ($N = 1608$) *n* (%)	Pre COVID-19 (3/18/2019–3/17/2020) ($n^* = 873$) *n* (%)	Post COVID-19 (3/18/2020–3/18/2021) ($n^* = 735$) *n* (%)	*p*
Age group				.002[a]
0–4	574 (35.6%)	333 (38.1%)	241 (32.8%)	
5–10	280 (17.4%)	170 (19.5%)	110 (15.0%)	
11–13	251 (15.6%)	118 (13.5%)	133 (18.1%)	
14–17	503 (31.2%)	252 (28.9%)	251 (34.1%)	
Sex				.13[b]
Female	851 (52.9%)	447 (51.2%)	404 (55.0%)	
Male	757 (47.1%)	426 (48.8%)	331 (45.0%)	
Geographic location				.30[b]
Urban	853 (53%)	473 (54.3%)	380 (51.7%)	
Non-urban	753 (46.8%)	398 (45.7%)	355 (48.3%)	
Race/ethnicity				.52[b]
White, Non-Hispanic (NH)	1320 (82%)	715 (82.5%)	605 (83.3%)	
Black, NH	158 (9.8%)	87 (10.0%)	71 (9.8%)	
Hispanic	92 (5.7%)	49 (5.6%)	43 (5.9%)	
Other	23 (1.4%)	16 (1.9%)	7 (1.0%)	

Note: Although there were 1981 and 2235 encounters in the pre- and post-periods, there were 873 and 735 unique patients in each, respectively.
[a] *p* from Mann-Whitney *U* test.
[b] *p* from Chi-square test of association.

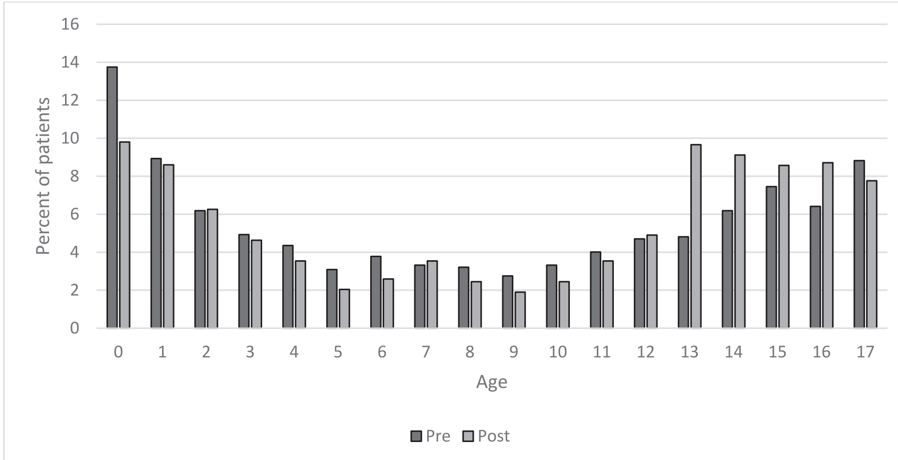

Fig. 1. Distribution of age among unique patients in the Pre and Post periods ($n = 873$ and $n = 735$ patients, respectively).

subsequent encounter and sequela, respectively. In the post-COVID-19 time period, 98.5% ($n = 1443$) of the encounters can be accounted for with these same codes.

The text below describes our analysis of age subgroup by specific ICD-10 code (e.g., T36). For those aged 11–13 years, there were significant increases in the rate of encounters due to *Non-opioid analgesics antipyretic and antirheumatic* (T39, $IRR = 2.25$, $p = .003$), *Psychotropic drugs* (T43, $IRR = 2.14$, $p = .003$) and *Diuretics and other unspecified drugs, medicants and biologic substances* (T50, $IRR = 3.01$, $p < .001$). There was a significant decrease in the rate of encounters due to *Narcotics and psychodysleptics* (T40, $IRR = 0.19$, $p < .001$). For the oldest group, ages 14–17, there were significant increases in the rate of encounters for five ICD-10 codes which included: *Hormones and their synthetic substitutes and antagonists* (T38, $IRR = 1.94$, $p = .02$), *Antiepileptic sedative, hypnotics and antiparkinsonism* (T42, $IRR = 2.15$, $p = .009$), *Psychotropic drugs* (T43, $IRR = 1.63$, $p = .006$), *Systemic and hematologic agents, not otherwise specified* (T50, $IRR = 1.92$, $p < .001$) and *Diuretics and other unspecified drugs, medicants and biologic substances* (T50, $IRR = 1.40$, $p = .047$). There were no significant decreases in any of the specific poisoning codes within this age group.

Discussion

This study characterizes pediatric poisoning before and after COVID-19. Important findings include a description of "drugs and medicants" implicated in the event, and identification of subgroups adversely affected (e.g., rural residents). Findings from this study suggest that there was significant shift in the ages of pediatric patients with poisoning between the year before COVID-19 (Pre) and the year immediately following (Post). While those aged 0–4 were less likely to experience poisoning in the Post-COVID-19 period, those aged 11–13 and 14–17 were more highly affected by poisoning in the Post- COVID-19 period.

Table 2
Comparison on pediatric poisoning rates overall and by demographic group by study period.

	Pre COVID-19 (3/18/2019–3/17/2020) (n = 321,979 encounters)		Post COVID-19 (3/18/2020–3/18/2021) (n = 257,674 encounters)		Incidence rate ratio *IRR*(95% CI)	*p*
	n	*IR*	*n*	*IR*		
Overall	1981	6.15	2235	8.67	1.41 (1.33, 1.50)	<0.001
Age group						
0–4	694	2.16	701	2.72	1.26 (1.14, 1.40)	<0.001
5–10	549	1.71	683	2.65	1.55 (1.39, 1.74)	
11–13	303	0.94	290	1.13	1.20 (1.02, 1.41)	<0.0010 .030
14–17	435	1.35	561	2.18	1.61 (1.42, 1.83)	<0.001
Sex						
Female	899	2.79	1045	4.06	1.45 (1.33, 1.59)	<0.001
Male	1082	3.36	1190	4.62	1.37 (1.27, 1.49)	<0.001
Geographic						
location	909	2.82	930	3.61	1.28 (1.17, 1.40)	<0.001
Urban	1070	3.32	1305	5.06	1.52 (1.41, 1.65)	<0.001
Non-urban						
Race/ethnicity						
White, Non-Hispanic	1717	5.33	2006	7.79	1.46 (1.37, 1.56)	<0.001
(NH)	122	0.38	105	0.41	1.08 (0.83, 1.40)	0.58
Black, NH	90	0.28	102	0.40	1.42 (1.07, 1.88)	0.016
Hispanic	46	0.14	13	0.05	0.35 (0.14, 0.64)	<0.001
Other						
Visit type						
Outpatient	1421	4.41	1599	6.21	1.41 (1.31, 1.51)	<0.001
Inpatient	560	1.74	636	2.47	1.42 (1.27, 1.59)	<0.001
Emergency	544	1.69	449	1.74	1.03 (0.91, 1.17)	0.63

Note: Incidence rates (*IR*) calculated per 1000 encounters.

Table 3
Comparison of pediatric poisoning claims by ICD10 code by study period.

ICD-10 code description Poisoning by, adverse effect of and underdosing of	Pre *n*	Post *n*	Incidence rate ratio *IRR* (95% CI)	*p*
T36 – Systemic antibiotics	159	63	0.50 (0.37, 0.66)	<0.001
T37 – Other systemic ant-infectives and antiparasitic	18	14	0.97 (0.47, 1.96)	0.94
T38 – Hormones and their synthetic substitutes and antagonists	61	72	1.48 (1.05, 2.08)	0.026
T39 – Non opioid analgesics antipyretic and antirheumatic	101	106	1.31 (0.99, 1.72)	0.052
T40 – Narcotics and pyschodysleptics	74	48	0.81 (0.56, 1.16)	0.26
T41 – Anesthetics and therapeutic gases	9	2	0.28 (0.04, 1.17)	0.085
T 42 – Antiepileptic sedative, hypnotics and antiparkinsonism	61	64	1.31 (0.92, 1.87)	0.13
T43 – Psychotropic drugs	112	132	1.47 (1.15, 1.90)	0.003
T44 – Agents primarily affecting the autonomic nervous system	22	16	0.91 (0.47, 1.73)	0.78
T45 – Systemic and hematologic agents, not otherwise specified	1100	1465	1.66 (1.54, 1.80)	<0.001
T46 – Agents primarily affecting the cardiovascular system	24	26	1.35 (0.77, 2.38)	0.29
T47 – Agents primarily affecting the gastrointestinal system	18	14	0.97 (0.47, 1.96)	0.94
T48 – Agents primarily acting on the smooth and skeletal muscle and the respiratory system	18	10	0.69 (0.31, 1.50)	0.36
T49 – Topical agents primarily affecting skin and mucous membranes and by ophthalmological, otorhinolaryngological and dental drug	14	14	1.25 (0.59, 2.66)	0.56
T50 – Diuretics and other unspecified drugs, medicants and biological substances	214	204	1.19 (0.98, 1.44)	0.074

Note: Pre-COVID-19 (3/18/2019–3/17/2020), $n = 321{,}979$ encounters; Post COVID-19 (3/18/2020–3/18/2021), $n = 257{,}674$ encounters.

This study identifies IRRs >1 in nearly every demographic subgroup with the exception of black, non-Hispanic, identifying increased incident rates in the Post-COVID-19 time period for nearly every demographic group included in our analysis.

The comparison of specific 'drugs or medicants', independently and then also by age group, identifies a significant decrease in poisoning by *Systemic antibiotics* (T36) which likely reflects the decrease in infectious conditions during the COVID-19 time period requiring treatment with antibiotics (Li et al., 2022). Important implications are that resumption of social interactions among children will likely result in Pre-COVID-19 rates and preventative measures should be taken to prevent child injury.

In contrast, this study's adolescents ages 14–17 as having significantly increased rates Post-COVID-19 for "drugs or medicants" that are frequently misused or abused (e.g., sedatives, narcotics, psychotropics) (Substance Abuse and Mental Health Services Administration, 2022). The *U.S. Department of Health and Human Services Substance Abuse and Mental Health Services Administration* identifies specific prescription drugs falling in the classification of stimulants, opioids, and depressants as frequently misused or abused by teens (Substance Abuse and Mental Health Services Administration, 2022). Similarly, Lelak et al. (2021) report findings consistent with these results: they report an increase in the number of pediatric poisoning in adolescents; their sample grouped this age from 13 to 19 years and they identified a 6% increase between time periods ($p < .001$). These findings are supported by the literature which reported increased drug overdose mortality in those aged 15–35 in the time period affected by COVID-19 (Lee & Singh, 2022). Clinically, healthcare providers need understand the effects of COVID-19 on drug use and in turn, screen and provide appropriate treatment for adolescents in need of services.

This study elucidates the effects of COVID-10 on poisoning by systemic and hematologic agents, which accounted for a significant proportion of encounters in the Pre- ($n = 1100$) and Post-COVID-19 ($n = 1465$). Our findings show the majority of these encounters were related to the adverse effects of antineoplastic and immunosuppressive drugs. It is possible the higher IRR specific to this code relates to our analytic approach which considers healthcare utilization. Specifically, other infectious diseases were less prevalent in the COVID-19 affected time period; as such, pediatric cancers accounted for a larger portion of health care services billed (DeLaroche et al., 2021; Hartnett et al., 2020). The increased acuity of patients in the time period affected by COVID-19 is supported in the literature about the effects of COVID-19 on pediatric poisonings (Lelak et al., 2021; Schultz et al., 2022).

Findings from this study showing that IRRs increased between Pre- and Post-COVID-19 in nearly every demographic subgroup both conflict and are supported by literature. One study reports a decrease in the absolute number of pediatric poisonings using an administrative dataset (Schultz et al., 2022). The discrepancy the Schulz study and this study's findings could be attributed to our use of IRRs, which account for changes in healthcare utilization before and during the COVID-19

Table 4
Comparison of pediatric poisoning claims by ICD-10 code by study period, stratified by age.

ICD-10 code description	Age 0–4		Age 5–10		Age 11–13		Age 14–17	
	IRR (95% CI)	*p*	IRR (95% CI)	*p*	IRR (95% CI)	*p*	IRR (95% CI)	*p*
T36 – Systemic antibiotics	**0.51 (0.34, 0.74)**	**<0.001**	**1.65 (1.47, 1.85)**	**<0.001**	0.29 (0.08, 0.83)	0.018	1.16 (0.54, 2.50)	0.70
T38 – Hormones and their synthetic substitutes and antagonists	**2.40 (1.24, 4.84)**	**0.009**	0.77 (0.30, 1.86)	0.57	0.67 (0.27, 1.56)	0.36	**1.94 (1.11, 3.45**	**0.020**
T39 – Non opioid analgesics antipyretic and antirheumatic	0.80 (0.29, 2.06)	0.65	1.11 (0.41, 2.95)	0.83	**2.25 (1.31, 3.95)**	**0.003**	1.13 (0.78, 1.62)	0.52
T40 – Narcotics and pyschodysleptics	1.17 (0.69, 1.96)	0.56			**0.19 (0.06, 0.49)**	**<0.001**	1.00 (0.46, 2.15)	0.99
T42 – Antiepileptic sedative, hypnotics and antiparkinsonism	0.83 (0.39, 1.76)	0.63	1.00 (0.46, 2.15)	0.99	1.13 (0.44, 2.83)	0.80	**2.15 (1.21, 3.92)**	**0.009**
T43 – Psychotropic drugs	0.66 (0.27, 1.56)	0.36	0.74 (0.32, 1.60)	0.45	**2.14 (1.30, 3.58)**	**0.003**	**1.63 (1.15, 2.32)**	**0.006**
T45 – Systemic and hematologic agents, not otherwise specified	**1.56 (1.36, 1.79)**	**<0.001**	**1.90 (1.67, 2.15)**	**<0.001**	1.07 (0.85, 1.35)	0.55	**1.92 (1.58, 2.33)**	**<0.001**
T50 – Diuretics and other unspecified drugs, medicants and biological substances	0.98 (0.71, 1.34)	0.90	**0.58 (0.34, 0.98)**	**0.042**	**3.01 (1.73, 5.43)**	**<0.001**	**1.40 (1.01, 1.96)**	**0.047**

Note: Pre-COVID-19 (3/18/2019–3/17/2020), $n = 321{,}979$ encounters; Post COVID-19 (3/18/2020–3/18/2021), $n = 257{,}674$ encounters.

pandemic. A comparison of absolute counts versus rates which consider that many nonemergent procedures were not permitted in many regions of the U.S., is an important consideration (Schultz et al., 2022). Additional studies based on U.S. Poison Control data also note a decrease in the time period affected by COVID-19; yet, there was an increased number of children who required hospitalization or intensive care management or the poisoning resulted in death (Ciccotti et al., 2022; Lelak et al., 2021). Additional studies report a decrease in pediatric emergency department visits for injuries including poisoning to decrease by 33% during the time period affected by COVID-19 (DeLaroche et al., 2021; Park et al., 2022). In support of our findings, Zhang et al. (2022) report the number of pediatric emergency department visits for poisonings as recorded in an injury database in Canada increased from 52/10,000 visits to 117/10,000 visits in the time period affected by COVID-19 ($p < .001$). An additional study reporting severe disease requiring intensive care in the United Kingdom rates of poisoning and other injuries were not significantly different between time periods ($p = .86$) (Williams et al., 2021). Differing results are likely due to the differing approaches to healthcare utilization and differing sources of data and groupings (e.g., grouping poisoning with injury).

Limitations and strengths

This study has unique strengths that contribute to the understanding of the effects of the COVID-19 pandemic had on pediatric poisonings. We identified that the incidence rate associated with pediatric poisoning encounters increased between the two time periods, and we also described how this overall increase was observed in nearly all demographic subgroups. To further elucidate the mechanisms of the increase we evaluated differences in the incidence of pharmacologic agents that resulted in pediatric poisoning. The findings from this study, using IRRs, also take into account the immediate decline in healthcare utilization following the emergency of COVID-19, which is critical to making an accurate comparison between these two highly distinctive time periods.

The limitations largely stem from the use of an existing dataset of billed encounters. These encounters could inaccurately capture patient intention (e.g., suicide ideation or attempt). This study does not infer causality and as such, it is possible that findings reflect a pre-existing trend unrelated to the effect of COVID-19. Similarly, there is variability in the application of the guideline for the use of *ICD-10-CM Official Guidelines for Coding and Reporting* in the context of poisoning by specific drugs or pharmacologic agents (Centers for Medicare and Medicaid Services, 2024). We acknowledge that since the number of encounters exceeds the number of unique patients in both time periods, the incidence rates may not capture the underlying prevalence of new cases in either time period. However, this concern is mitigated since our primary focus is on the incidence rate ratios, and repeated visits by at least some of the patient population is a feature that is present in both time periods. Finally, these data reflect one health care system, which limits generalization to a broader patient population.

Implications to practice

Findings from this study contribute to nursing practice. Specially, nurses are instrumental to the prevention and treatment of pediatric injuries such as pediatric poisonings (Bernardo, 2002). Because the implications of COVID-19 persist despite the dissipation of effects on the daily living of individuals (e.g., increase in mental health issues), it is critically important that nurses recognize the implications of COVID-19 on pediatric poisonings (Substance Abuse and Mental Health Services Administration, 2023). Of great importance, nurses must recognize those at high-risk for pediatric poisoning. As such, this study, which identifies highly affected pediatrics populations, can guide the care the nurses provide to patients. Of the high-risk populations, one of the most notable identified in this study was the adolescent population. The drugs or medicants ingested by adolescents were often those associated with substance misuse; as such, nurses need to implement recommended guidelines for adolescent screening and when substance use disorders are identified, facilitating medical management (Substance Abuse and Mental Health Services Administration, 2022).

Conclusion

The effects of the pandemic on pediatric poisoning, specifically evaluating the effects of rural residence, have not been fully characterized. This article contributes to this understanding by describing relative risk of poisoning among demographic subgroups as well as relative risk of ingestion of specific agents within four pediatric age categories.

Data sharing agreement

Deidentified data was provided by the University of Kentucky Center for Clinical and Translational Science Honest Broker Agreement.

CRediT authorship contribution statement

Elizabeth Salt: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Amanda Thaxton Wiggins:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Christina Howard:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Gena L. Cooper:** Writing – review & editing, Writing – original draft, Conceptualization. **Tom C. Badgett:** Writing – original draft, Validation, Conceptualization. **Kara Rasheed:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Emily McSween:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Mary Kay Rayens:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

Authors do not have any interests to report related to this work.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pedn.2024.07.006.

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